

Audit S3 Buckets for Public Access and Notify

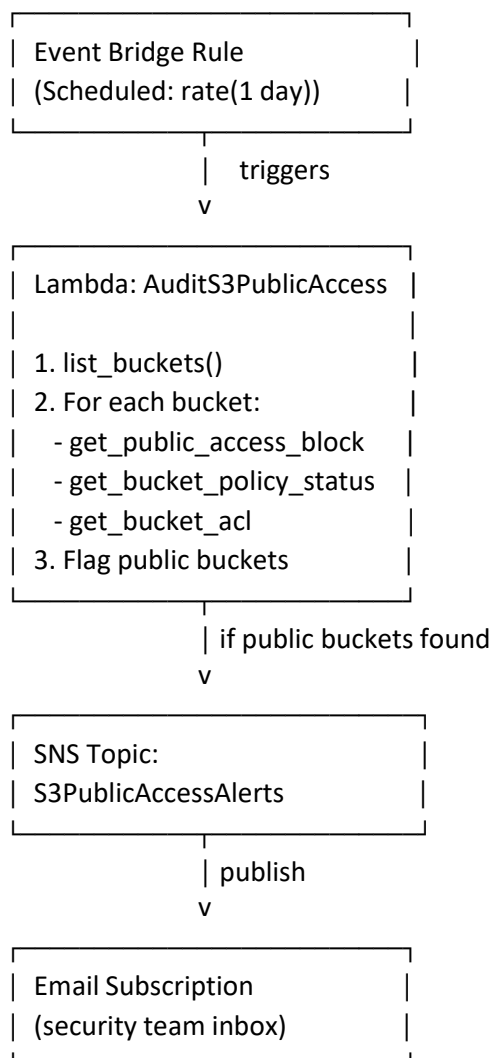
Friday, July 17, 2026 5:27 PM

Task 6: Audit S3 Buckets for Public Access and Notify

Objective

Detect any S3 bucket that is publicly accessible and alert via SNS — checking **Block Public Access (BPA) configuration**, **bucket policy status**, and **ACL grants**, since new buckets have BPA enabled and ACLs disabled by default.

Architecture Diagram:

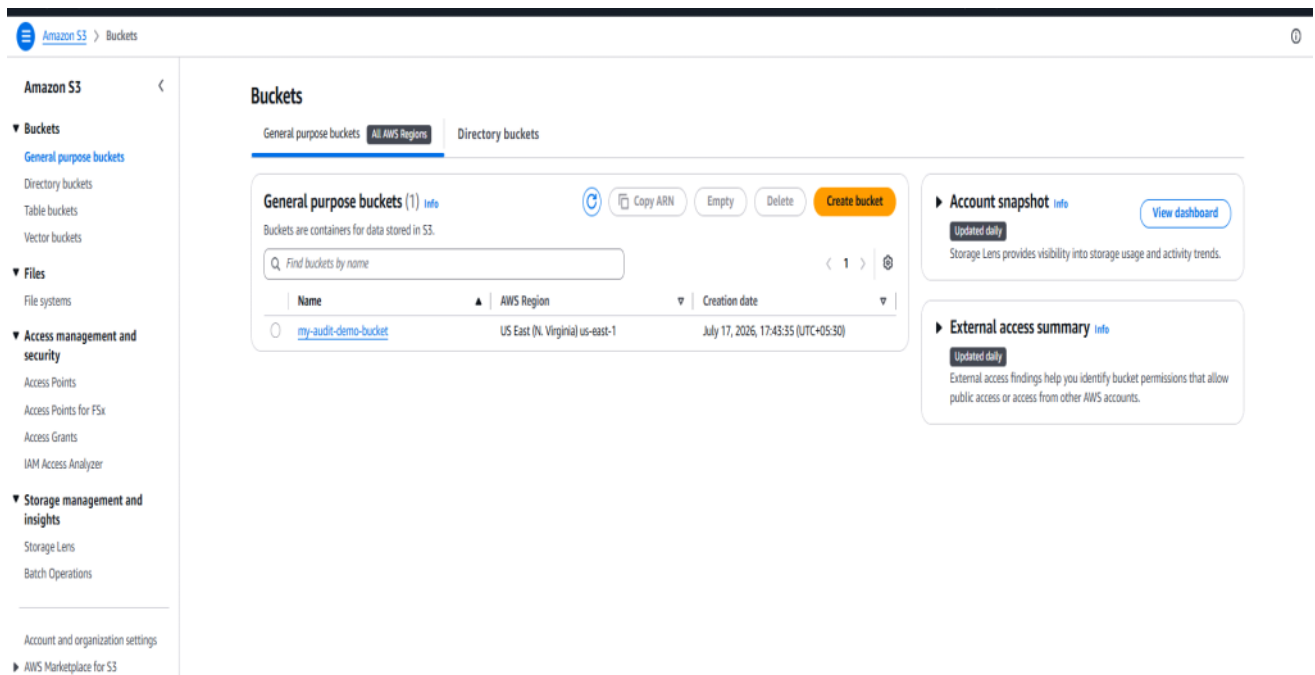


Prerequisites

- AWS CLI configured (aws configure) with permissions to create SNS topics, IAM roles, Lambda functions, and EventBridge rules.
- At least one S3 bucket in the account to audit (for meaningful test results).

CLI:

```
PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task6\S3-Public-Alert> aws s3 mb
s3://my-audit-demo-bucket --region us-east-1
make_bucket: my-audit-demo-bucket
PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task6\S3-Public-Alert> aws s3 ls
2026-07-17 17:43:35 my-audit-demo-bucket
```



```
# Create test files
echo "test file 1" > file1.txt
echo "test file 2" > file2.txt
echo "test file 3" > file3.txt
echo "test file 3" > file4.txt

# Upload
aws s3 cp file1.txt s3://my-cleanup-demo-bucket/
aws s3 cp file2.txt s3://my-cleanup-demo-bucket/
aws s3 cp file3.txt s3://my-cleanup-demo-bucket/
aws s3 cp file4.txt s3://my-cleanup-demo-bucket/
```

```
PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task6\S3-Public-Alert> aws s3 ls s3://my-
audit-demo-bucket/
2026-07-17 17:55:36      13 file1.txt
2026-07-17 17:55:42      13 file2.txt
2026-07-17 17:55:48      13 file3.txt
PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task6\S3-Public-Alert>
```

Step 1 — SNS Setup (Topic + Email Subscription)

Console steps (detailed navigation)

1. Open the **AWS Console** → search for and open **SNS**.
2. In the left sidebar, click **Topics**.

3. Click **Create topic** (top right).
4. **Type**: select **Standard**.
5. **Name**: S3PublicAccessAlerts.
6. (Optional) **Display name**: S3-Public-Alert.
7. Leave other settings default → click **Create topic**.
8. On the topic's detail page, click **Create subscription**.
9. **Protocol**: select **Email**.
10. **Endpoint**: enter your email address (e.g. you@example.com).
11. Click **Create subscription**.
12. Check your email inbox for a message titled "**AWS Notification - Subscription Confirmation**" → click **Confirm subscription** in that email.
13. Back in the SNS console, refresh the **Subscriptions** tab and confirm status shows Confirmed.
14. Copy the **Topic ARN** from the topic detail page (top of page) — you'll need it for the Lambda environment variable in Step 3.

Amazon SNS > Topics > Create topic

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create topic

Details

Type [info](#)
Topic type cannot be modified after topic is created

☐ **FIFO (first-in, first-out)**

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ **Standard**

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [info](#)
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

Maximum 100 characters.

► **Encryption - optional**
Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

► **Access policy - optional** [info](#)
This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

Amazon SNS > Topics > S3PublicAccessAlerts

0

Topic S3PublicAccessAlerts created successfully.
You can create subscriptions and send messages to them from this topic.

Notifications 0 0 0 1 0 0

Publish message

S3PublicAccessAlerts

Edit Delete Publish message

Details

NameS3PublicAccessAlerts

ARNarn:aws:sns:us-east-1:675134942081:S3PublicAccessAlerts

Display nameS3 Public Access Alerts

TypeStandard

Topic owner675134942081

Subscriptions

Access policy Delivery policy (HTTP/S) Delivery status logging Encryption Tags Integrations

Subscriptions (0)

Edit Delete Request confirmation Confirm subscription Create subscription

Q Search

< 1 >

ID	Endpoint	Status	Protocol
No subscriptions found			
You don't have any subscriptions to this topic.			

Create subscription

Amazon SNS > Subscriptions > Create subscription

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create subscription

Details

Topic ARNarn:aws:sns:us-east-1:675134942081:S3PublicAccessAlerts

ProtocolThe type of endpoint to subscribeEmail

EndpointAn email address that can receive notifications from Amazon SNS.suhascor@gmail.com

After your subscription is created, you must confirm it. [Info](#)

Subscription filter policy - optional [Info](#)

This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional [Info](#)

Send undeliverable messages to a dead-letter queue.

Cancel Create subscription

Amazon SNS > Topics > S3PublicAccessAlerts

Amazon SNS

- Dashboard
- Topics
- Subscriptions
- Mobile
 - Push notifications
 - Text messaging (SMS)

S3PublicAccessAlerts Edit Delete Publish message

Details

Name S3PublicAccessAlerts	ARN arn:aws:sns:us-east-1:675134942081:S3PublicAccessAlerts	Display name S3 Public Access Alerts	Type Standard
Topic owner 675134942081			

Subscriptions | Access policy | Delivery policy (HTTP/S) | Delivery status logging | Encryption | Tags | Integrations

Subscriptions (1) Edit Delete Request confirmation Confirm subscription Create subscription

Q Search

ID	Endpoint	Status	Protocol
1b56017b-47fe-468c-a55a-0abb1e7a96ae	suhascore@gmail.com	Confirmed	EMAIL

Step 2 — IAM Role for Lambda Console steps (detailed navigation)

Create the custom policy:

1. Open **IAM** → left sidebar → **Policies**.
2. Click **Create policy** (top right).
3. Click the **JSON** tab → delete placeholder → paste the policy JSON above (with your real SNS ARN).
4. Click **Next**.
5. **Policy name:** S3AuditPolicy.
6. Click **Create policy**.

IAM > Policies

Identity and Access Management (IAM)

Q Search IAM

Dashboard

Access Management

- Roles
- Policies**
- IAM users
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
- Resource analyzer

Policies Info

A policy is an object in AWS that defines permissions.

Q Search

Filter by Type
All types

Policy name	Type	Used as	Description
Loading policies			

Actions Delete Create policy

IAM > Policies > Create policy

Step 2
Review and create

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "S3AuditPermissions",
6       "Effect": "Allow",
7       "Action": [
8         "s3:ListAllMyBuckets",
9         "s3:GetBucketPublicAccessBlock",
10        "s3:GetBucketPolicyStatus",
11        "s3:GetBucketAcl",
12      ],
13      "Resource": "*"
14    },
15    {
16      "Sid": "SNSPublishPermission",
17      "Effect": "Allow",
18      "Action": "sns:Publish",
19      "Resource": "arn:aws:sns:us-east-1:675134842881:S3PublicAccessAlerts"
20    }
21  ]
22 }

```

Visual JSON Actions

Edit statement
SNSPublishPermission Remove

Add actions

Choose a service

Q Search services

Included

SNS

Available

AI Operations

AMP

API Gateway

API Gateway V2

ARC Region switch

ARC Zonal Shift

ASC

Access Analyzer

Add a resource Add

Add a condition - optional Add

19/75 JSON

795 of 6144 characters remaining

Security: 0 Errors: 0 Warnings: 0 Suggestions: 0

Cancel Next

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

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Unused access

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Organization activity

Service control policies

Resource control policies

Policy S3AuditPolicy created.

Policies (1532) info

A policy is an object in AWS that defines permissions.

Filter by Type

Customer managed 9 matches

Policy name	Type	Used as
Amazon.EventBridge_Invoke_Lambda_1557282200	Customer managed	Permissions policy (1)
Amazon.EventBridge_Invoke_Lambda_635542110	Customer managed	Permissions policy (1)
AWSLambdaBasicExecutionRole-13a36d8b-aea3-4821-8c09-8f64f8b63bae	Customer managed	Permissions policy (1)
AWSLambdaBasicExecutionRole-174c3ac6-06c0-4092-a1d5-3b3c0fd96046	Customer managed	Permissions policy (1)
AWSLambdaBasicExecutionRole-43007d51-3862-476d-bf76-496374ae1f2c	Customer managed	Permissions policy (1)
AWSLambdaBasicExecutionRole-5730fbc6-4262-4842-b9e7-4c4421dd4f1c	Customer managed	Permissions policy (1)
AWSLambdaBasicExecutionRole-933a4c0b-4aa3-4934-9804-3ad1d8403d41	Customer managed	Permissions policy (1)
AWSLambdaBasicExecutionRole-c03af1a1-a1d7-4c7-84ba-eca4f05d8b0b5	Customer managed	Permissions policy (1)
S3AuditPolicy	Customer managed	None

Create the role:

- Left sidebar → Roles → Create role.
- Trusted entity type: AWS service.
- Use case: Lambda → Next.
- Search and check S3AuditPolicy.
- Search and check AWSLambdaBasicExecutionRole.
- Click Next.
- Role name: S3AuditLambdaRole.
- Click Create role.
- Open the role and copy its ARN (needed in Step 3).

- Step 1
● Select trusted entity
- Step 2
○ Add permissions
- Step 3
○ Name, review, and create

Select trusted entity [info](#)

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Lambda

Choose a use case for the specified service.

Use case

- ☒ **Lambda**
Allows Lambda functions to call AWS services on your behalf.
- ☐ **AWSServiceRoleForLambda**
Allows Lambda to manage AWS resources on your behalf.

Cancel **Next**

- Step 1
● Select trusted entity
- Step 2
● Add permissions
- Step 3
○ Name, review, and create

Add permissions [info](#)☒ Use existing policy

Attach an existing managed policy to the role.

☐ Create inline policy

Create a new policy that will be attached directly to the role.

Permissions policies (1/1185) [info](#)

Choose one or more policies to attach to your new role.

Q Search		Filter by Type	Customer managed 9 matches		< 1 > ⓘ
☒ Policy name ⓘ	Type			Description	
☐ Amazon_EventsBridge_Invoke_Lambda_1557282100	Customer managed			-	
☐ Amazon_EventsBridge_Invoke_Lambda_635542110	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-13a36d88-aea3-4821-8cd9-8fb4fb8633e	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-174c3ac6-06c0-4692-a1d5-5b3cbf996046	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-43007d31-3862-476d-bf76-4f6574ae1f2c	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-5730fb6f-f2d2-4842-b9d7-4cf42c1ddff1c	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-939a4dbb-4ea3-4934-9804-9ad18f40b41	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-cd0af1a1-a1d7-4cc7-84ba-ecaaf05d8b0d6	Customer managed			-	
☒ SSIAuditPolicy	Customer managed			-	

► Set permissions boundary - optional

Cancel **Previous** **Next**

- Step 1
● Select trusted entity
- Step 2
● Add permissions
- Step 3
○ Name, review, and create

Add permissions [info](#)☒ Use existing policy

Attach an existing managed policy to the role.

☐ Create inline policy

Create a new policy that will be attached directly to the role.

Permissions policies (1/1185) [info](#)

Choose one or more policies to attach to your new role.

Q AWSLambdaBasicExecutionRole		Filter by Type	All types 7 matches		< 1 > ⓘ
☒ Policy name ⓘ	Type			Description	
☒ AWSLambdaBasicExecutionRole	AWS managed			Provides write permissions to CloudWatc...	
☐ AWSLambdaBasicExecutionRole-13a36d88-aea3-4821-8cd9-8fb4fb8633e	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-174c3ac6-06c0-4692-a1d5-5b3cbf996046	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-43007d31-3862-476d-bf76-4f6574ae1f2c	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-5730fb6f-f2d2-4842-b9d7-4cf42c1ddff1c	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-939a4dbb-4ea3-4934-9804-9ad18f40b41	Customer managed			-	
☐ AWSLambdaBasicExecutionRole-cd0af1a1-a1d7-4cc7-84ba-ecaaf05d8b0d6	Customer managed			-	

► Set permissions boundary - optional

Cancel **Previous** **Next**

- Step 1
● Select trusted entity
- Step 2
● Add permissions
- Step 3
● **Name, review, and create**

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

S3AuditLambdaRole

Maximum 64 characters. Use alphanumeric and "+, @, _" characters.

Description

Add a short explanation for this role.

Allows Lambda functions to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: "_", "+", "@", "/", "()", "[]", "%", ":", "=".

Step 1: Select trusted entities

Edit

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "lambda.amazonaws.com"
12        ]
13      }
14    ]
15  }
16 }
```

Step 2: Add permissions

Edit

16

Step 2: Add permissions

Edit

Permissions policy summary

Policy name ↕

▲

Type

▼

Attached as

▼

[AWSLambdaBasicExecutionRole](#)

AWS managed

Permissions policy

[S3AuditPolicy](#)

Customer managed

Permissions policy

Step 3: Add tags

Identity and Access Management (IAM)

Q Search IAM

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Resource control policies

Role deleted s3-cleanup-lambda-role.

Roles (19) info

C

Delete

Create role

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Q Search

< 1 > ⚙

☐	Role name ▲	Trusted entities	Last activity ▼
☐	demoFunction-role-ru7ank3u	AWS Service: lambda	-
☐	Lambda_Demo-role-c2ox3umo	AWS Service: lambda	26 days ago
☐	layer-lambda-demo-role-fu6markg	AWS Service: lambda	26 days ago
☐	new_function-role-b63f609	AWS Service: lambda	22 days ago
☐	new_function-role-j5ilwae5	AWS Service: lambda	22 days ago
☐	new_function-role-wfayjaut	AWS Service: lambda	24 days ago
☐	new_function-role-ygsvues	AWS Service: lambda	-
☐	OrderProcessorLambdaRole	AWS Service: lambda	-
☐	OrderProcessorLambdaRole_SQS_SNS	AWS Service: lambda	21 days ago
☐	S3AuditLambdaRole	AWS Service: lambda	-

Roles Anywhere info

Manage

Authenticate your non AWS workloads and securely provide access to AWS services.

Identity and Access Management (IAM)

Permissions policies (2) info

You can attach up to 10 managed policies.

Filter by Type

Search

All types

Policy name	Type	Attached entities
AWSLambdaBasicExecutionRole	AWS managed	2
S3AuditPolicy	Customer managed	1

S3AuditPolicy

Copy JSON Edit

```

3  "Statement": [
4    {
5      "Sid": "S3AuditPermissions",
6      "Effect": "Allow",
7      "Action": [
8        "s3:ListAllMyBuckets",
9        "s3:GetBucketPublicAccessBlock",
10       "s3:GetBucketPolicyStatus",
11       "s3:GetBucketAcl"
12     ],
13     "Resource": "*"
14   },
15   {
16     "Sid": "SNSPublishPermission",
17     "Effect": "Allow",
18     "Action": "sns:Publish",
19     "Resource": "arn:aws:sns:us-east-1:675134942081:S3PublicAccessAlerts" add your SNS ARN
20   }
21 ]
22

```

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a managed policy based on the service activities for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the actions and actions used and generate a policy. Learn more.

Step 3 — Lambda Function (Boto3)

Console steps (detailed navigation)

1. Open **Lambda** → **Create function**.
2. Select **Author from scratch**.
3. **Function name**: AuditS3PublicAccess.
4. **Runtime**: Python 3.12.
5. Expand **Change default execution role** → **Use an existing role** → select S3AuditLambdaRole.
6. Click **Create function**.
7. In **Code source**, replace the boilerplate with the code above → click **Deploy**.
8. Go to **Configuration** tab → **Environment variables** → **Edit**.
9. Click **Add environment variable**: Key = SNS_TOPIC_ARN, Value = your topic ARN from Step 1.
10. Click **Save**.
11. Still in **Configuration** → **General configuration** → **Edit** → set **Timeout** to 1 min 0 sec (listing/checking many buckets can take longer than the 3-second default) → **Save**.

→ [Lambda](#) → [Functions](#) → [Create function](#)

Create function info

Choose one of the following options to create your function.

☒ **Author from scratch**
Start with a simple Hello World example.

☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.

AuditS3PublicAccess

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime info
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.12

Permissions
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution role** under **Additional settings**.

Custom settings info

☒ **Durable execution**
Build resilient, stateful applications with automatic failure recovery.

☐ **EC2 capacity provider**
Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).

Custom settings Info

Durable execution
Build resilient, stateful applications with automatic failure recovery.

EC2 capacity provider
Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).

Additional settings
Customize your function architecture, execution role, HTTP(S) endpoints, tenant isolation mode, VPC, code signing, KMS key, and tags.

Cannot add or remove after creation

Cancel
Create function

Custom settings Info

Durable execution
Build resilient, stateful applications with automatic failure recovery.

EC2 capacity provider
Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).

Additional settings
Customize your function architecture, execution role, HTTP(S) endpoints, tenant isolation mode, VPC, code signing, KMS key, and tags.

General Info

ARM64 architecture
Use ARM64 instead of x86_64 (default).

Custom execution role
IAM permissions
[S3AuditLambdaRole](#)

Networking Info

Code
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General configuration
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Permissions
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Function URL
Environment variables
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VPC

Environment variables (1)

The environment variables below are encrypted at rest with the default Lambda service key.

Key	Value
SNS_TOPIC_ARN	arn:aws:sns:us-east-1:675134942081:S3PublicAccessAlerts

Step 4 — EventBridge: Schedule Daily

Console steps (detailed navigation)

1. Open **Amazon EventBridge** → left sidebar → **Rules**.
2. Click **Create rule**.
3. **Name**: DailyS3PublicAccessAudit.
4. **Event bus**: default.
5. **Rule type**: select **Schedule**.
6. Click **Next**.
7. **Schedule pattern**: select **A schedule that runs at a regular rate**.
8. Set **Rate expression**: 1 **Day(s)** (or use a Cron expression for a specific time, e.g. cron(0 8 * * ? *) for 8 AM UTC daily).
9. Click **Next**.

10. Target type: AWS service.
11. Select a target: Lambda function.
12. Function: select AuditS3PublicAccess.
13. Click **Next** → review → **Create rule**.

Amazon EventBridge > Scheduled rules

Dashboard

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Buses

- Event buses
- Rules **Updated**
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- Replays

Pipes

- Pipes

Scheduler

- Schedules
- Schedule groups
- Scheduled rules (legacy)** **Updated**

Integration

- Partner event sources
- API destinations
- Connections

Scheduled rules

Find scheduled rules

Any status

No scheduled rules
No scheduled rules to display.

Create scheduled rule

Amazon EventBridge > Scheduled rules > Create rule

Dashboard

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- Event buses
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- Pipes

Scheduler

- Schedules
- Schedule groups
- Scheduled rules (legacy)** **Updated**

Integration

- Partner event sources
- API destinations
- Connections

Schema registry

- Schemas

Step 1: Define rule detail

Step 2: Define schedule

Step 3: Select target(s)

Step 4 - optional: Configure tags

Step 5: Review and create

Define rule detail

Scheduled rule detail

Name

DailyS3PublicAccessAudit

Description - optional

Enter description

Event bus

default

Enable the rule on the selected event bus

EventBridge Scheduler - A new AWS scheduling capability!

A new EventBridge scheduling functionality that provides one-time and recurring scheduling functionality independent of Event buses and rules. You can create a schedule to invoke targets such as a Lambda function. [Learn More](#)

Continue in EventBridge Scheduler

Cancel Next

Step 1

Define rule detail

Step 2

Define schedule

Step 3

Select target(s)

Step 4 - optional

Configure tags

Step 5

Review and create

Define schedule

info

Schedule pattern

Schedule pattern

Choose the schedule type that best meets your needs.

☐ A fine-grained schedule that runs at a specific time, such as 8:00 a.m. PST on the first Monday of every month.

☒ A schedule that runs at a regular rate, such as every 10 minutes.

Rate expression

info

Enter a value and the unit of time to run the schedule.

Value

Unit, e.g. mins, hours...

Cancel

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(legacy)

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Free virtual workshops

New

Step 1

Define rule detail

Step 2

Define schedule

Step 3

Select target(s)

Step 4 - optional

Configure tags

Step 5

Review and create

Permissions

Note: When using the EventBridge console, EventBridge will automatically configure the proper permissions for the selected targets. If you're using the AWS CLI SDK, or CloudFormation, you'll need to configure the proper permissions.

Target 1

Target types

Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus

☐ EventBridge API destination

☒ AWS service

Select a target

info

Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

Lambda function

Target location

☒ Target in this account

☐ Target in another AWS account

Function

AuditS3PublicAccess

Configure version/alias

Permissions

☒ Use execution role (recommended)

Execution role

EventBridge needs permission to send events to the target specified above. By continuing, you are allowing us to do so. EventBridge and AWS Identity and Access Management

☐ Create a new role for this specific resource

☒ Use existing role

Role name

S3AuditLambdaRole

Additional settings

Add another target

Cancel

Skip to Review and create

Previous

Next

Step 5 — Testing

⚠ This step intentionally exposes a bucket publicly. Use a disposable test bucket with no sensitive data, and re-secure it immediately after confirming the alert.

Console steps (detailed navigation)

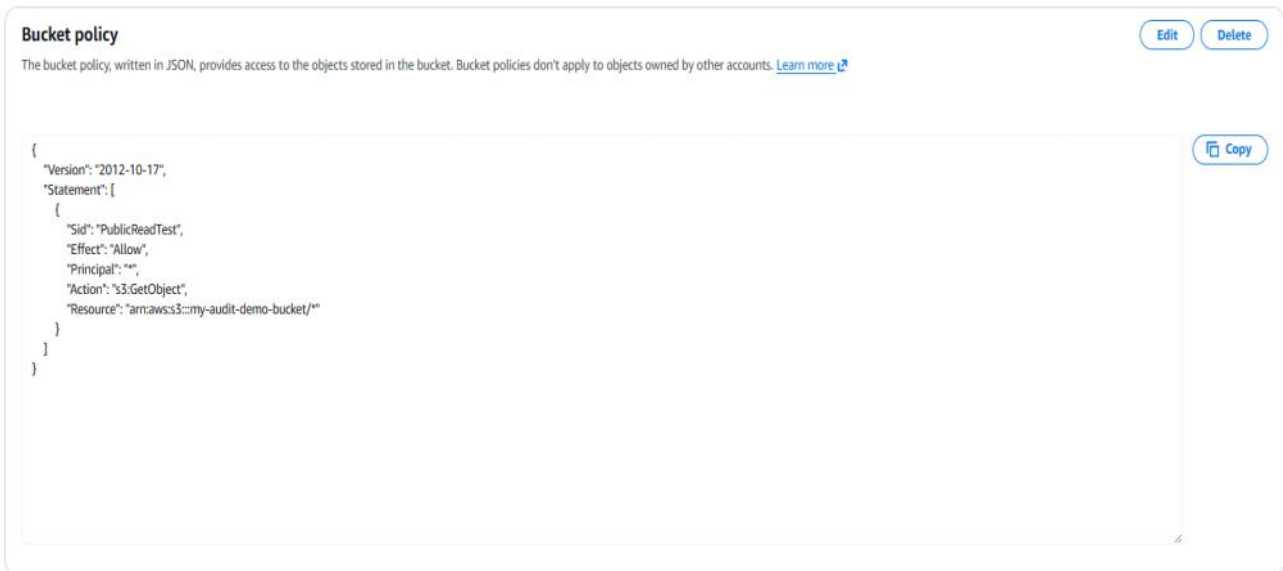
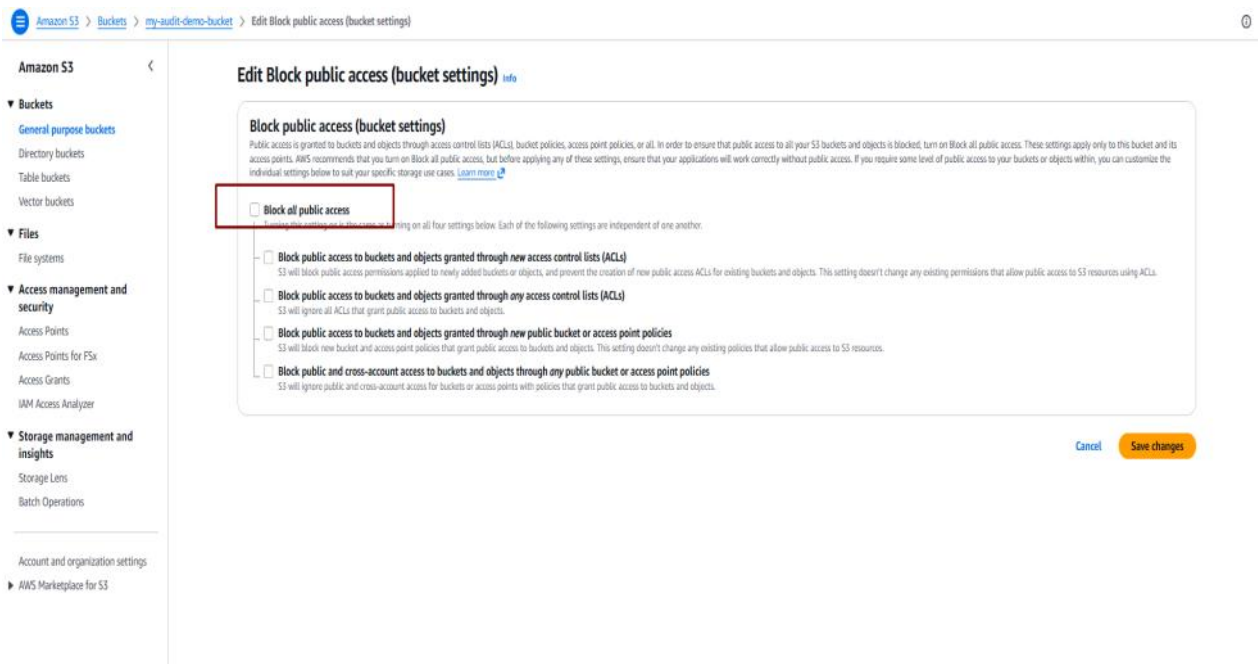
Create/select a test bucket:

1. Open **S3** → **Create bucket** (or select an existing empty test bucket).
2. **Bucket name**: e.g. my-test-public-audit-bucket-**<random>**.
3. Leave **Block all public access** checked for now → **Create bucket**.

Disable Block Public Access:

4. Open the test bucket → **Permissions** tab.

5. Under **Block public access (bucket settings)**, click **Edit**.
 6. Uncheck **Block all public access**.
 7. Type confirm in the field → click **Confirm**.
- Attach a public-read bucket policy:**
8. Still on the **Permissions** tab, scroll to **Bucket policy** → click **Edit**.
 9. Paste:



Run the Lambda manually:

11. Open **Lambda** → **AuditS3PublicAccess** → **Test** tab.
12. Create a test event (empty JSON {} is fine, the function doesn't require input) → click **Test**.
13. Check the execution results / **Monitor** → **View CloudWatch logs** — confirm the test bucket appears in the public_buckets list.
14. Check your email inbox for the SNS alert listing the bucket and the reason(s) it was flagged.

Log:

```
{
  "statusCode": 200,
  "public_bucket_count": 1,
  "public_buckets": [
    "my-audit-demo-bucket"
  ]
}
```

START RequestId: 26e347ee-0293-4371-9d4f-679fe831f000 Version: \$LATEST

Auditing 1 bucket(s)...

The following S3 bucket(s) are PUBLICLY ACCESSIBLE:

- my-audit-demo-bucket
 - Block Public Access is NOT fully enabled
 - Bucket policy evaluates as PUBLIC (IsPublic=True)

END RequestId: 26e347ee-0293-4371-9d4f-679fe831f000

REPORT RequestId: 26e347ee-0293-4371-9d4f-679fe831f000 Duration: 689.13 ms Billed Duration: 1038 ms Memory Size: 128 MB Max Memory Used: 98 MB Init Duration: 347.94 ms

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```
START RequestId: 26e347ee-0293-4371-9d4f-679fe831f000 Version: $LATEST
Auditing 1 bucket(s)...
The following S3 bucket(s) are PUBLICLY ACCESSIBLE:
• my-audit-demo-bucket
  - Block Public Access is NOT fully enabled
  - Bucket policy evaluates as PUBLIC (IsPublic=True)
END RequestId: 26e347ee-0293-4371-9d4f-679fe831f000
REPORT RequestId: 26e347ee-0293-4371-9d4f-679fe831f000    Duration: 689.13 ms    Billed Duration: 1038 ms    Memory Size: 128 MB    Max Memory Used: 98 MB    Init Duration: 347.94 ms
```

S3 Public Access Alert Inbox x



S3 Public Access Alerts <no-reply@sns.amazonaws.com>
to me ▾

7:13 PM (0 m)

The following S3 bucket(s) are PUBLICLY ACCESSIBLE:

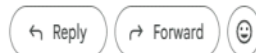
- my-audit-demo-bucket
 - Block Public Access is NOT fully enabled
 - Bucket policy evaluates as PUBLIC (IsPublic=True)

--

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<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:675134942081:S3PublicAccessAlerts:1b56017b-477e-468c-a55a-0abb1e7a96ae&Endpoint=suhascore@gmail.com>

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at <https://aws.amazon.com/support>



Re-secure the test bucket immediately after confirming the alert

Console:

1. Test bucket → **Permissions** → **Bucket policy** → **Edit** → delete the policy JSON → **Save changes**.
2. **Permissions** → **Block public access (bucket settings)** → **Edit** → check **Block all public access** → type confirm → **Confirm**.
3. (Optional) Delete the test bucket entirely: **S3** → select bucket → **Delete** → type the bucket name to confirm.

Edit Block public access (bucket settings) Info

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

[Cancel](#)

[Save changes](#)

Bucket policy

Edit

Delete

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Public access is blocked because Block Public Access settings are turned on for this bucket

To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about [using Amazon S3 Block Public Access](#)

No policy to display.

Copy

Amazon EventBridge

Scheduled rules

DailyS3PublicAccessAudit

Dashboard

Developer resources

Event buses

Pipes

Scheduler

Rule DailyS3PublicAccessAudit was updated successfully

DailyS3PublicAccessAudit

Edit

Disable

Delete

CloudFormation Template

Rule details

Rule name

DailyS3PublicAccessAudit

Description

-

Status

Enabled

Rule ARN

arn:aws:events:us-east-1:675134942081:rule/DailyS3PublicAccessAudit

Event bus name

default

Event bus ARN

arn:aws:events:us-east-1:675134942081:event-bus/default

Type

Scheduled Standard

Event schedule

Targets

Monitoring

Tags

Event schedule

Fixed rate of

1 minute

Edit

Triggers (1)

Info

Fix errors

Edit

Delete

Add trigger

Search for triggers

1

Trigger

EventBridge (CloudWatch Events): DailyS3PublicAccessAudit

arn:aws:events:us-east-1:675134942081:rule/DailyS3PublicAccessAudit

Rule state: ENABLED

Details

Description: None

Event bus: default

isComplexStatement: No

Schedule expression: rate(5 minutes)

Service principal: events.amazonaws.com

Statement ID: lambda-49147c7b-3289-4cca-89c2-6f326ef4817b

Accordingly Scheduled every 1 minutes execute and received the email Notification



S3 Public Access Alerts
The following S3 bucket(s) are PUBLICLY ACCESSIBLE: • my-audit-demo-bucket - Block Public Access is NOT fully enabled - Bucket policy evaluates as PUBLIC (IsPub

7:48 PM (3 minutes ago)



S3 Public Access Alerts <no-reply@sns.amazonaws.com>
to me
xxx

7:49 PM (2 minutes ago) ☆ ☹ ↶



S3 Public Access Alerts <no-reply@sns.amazonaws.com>
to me
xxx

7:50 PM (1 minute ago) ☆ ☹ ↶



S3 Public Access Alerts <no-reply@sns.amazonaws.com>
to me
xxx

The following S3 bucket(s) are PUBLICLY ACCESSIBLE:

• my-audit-demo-bucket
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- Bucket policy evaluates as PUBLIC (IsPublic=True)

-

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7:51 PM (0 minutes ago) ☆ ☹ ↶

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